THE EMERGENT-HOLONOMY COUNT IS A DEGREE-ZERO INVARIANT: $h(s) = \dim_k \mathrm{Hom}_{kU}(k[U/A], k[U/B])$

MACBETH

Abstract. When two agents composed as a Zappa–Szép (exact) factorisation act on a shared state and neither reserves the shared resource, an emergent holonomy appears: the two factor-orbits through a state s cross in $h(s)$ points, and $h(s) = 1$ exactly when the composition is aligned. We identify this count with a representation-theoretic invariant of the stabiliser group $U = \mathrm{Stab}_G(s)$. Writing $A, B \leq U$ for the two factor-isotropies and $k[U/A], k[U/B]$ for the associated permutation modules, the Mackey–Eckmann–Shapiro decomposition of Ext between permutation modules gives $h(s) = \dim_k \mathrm{Hom}_{kU}(k[U/A], k[U/B]) = \dim_k \mathrm{Ext}^0_{kU}(k[U/A], k[U/B])$. The novelty is not this classical formula but its consequence in the holonomy setting: transversality of an exact factorisation forces every relevant subgroup intersection to be trivial, so the entire higher Ext-tower vanishes identically. The holonomy invariant is therefore cohomological but concentrated in degree zero — a single Hom-space, or equivalently a double-coset count — with a theorem guaranteeing that no higher Ext obstruction can hide it. This corrects the working expectation that alignment should be detected by a higher cohomology class, and explains why an orientation-twist of the modules leaves the invariant unchanged. We close by placing this vanishing inside a single formula. Over an *abelian* ambient group the Mackey decomposition factors as $\mathrm{Ext}^n_{kG}(k[G/A], k[G/B]) \cong \mathrm{H}^n(A \cap B; k)^{\oplus [G:AB]}$: a degree-independent meeting count $h = [G : AB]$ times the cohomology of the overlap $A \cap B$. The holonomy tower $[h, 0, 0, \dots]$ is the transverse corner $A \cap B = \{e\}$ of this formula; a V_4 computation exhibits its full range, and support-variety geometry reads the dichotomy as an intersection of rank varieties — disjoint (transverse, vanishing) versus coincident (non-transverse, surviving).

1. Introduction

Orchestration and emergent holonomy. When two systems are composed, the order in which they act can matter. In the compositional theory of directed containers, the sharpest instance of this is the *Zappa–Szép* (or *exact*) factorisation of a group: $G = P \cdot P'$ with $P \cap P' = \{e\}$ and $|P||P'| = |G|$, so that every $g \in G$ has a unique expression $g = p\,p'$ and a unique reverse expression $g = p''p'''$. This is the algebraic skeleton of "two agents sharing one state, each with its own private moves," and it is the model we use throughout the container programme for agent orchestration, supply chains, and stateful smart contracts; that composing two directed containers this way is a Zappa–Szép product is due to Ahman–Uustalu [7], and the emergent-holonomy content below sits on top of their construction.

Fix a G-set S and a state $s \in S$. The two factors P, P' each sweep out an orbit through s, namely $P \cdot s$ and $P' \cdot s$. Composing the agents means asking how these two orbits interact, and the answer is a local invariant we call the *emergent holonomy*

$$h(s) \;:=\; \big|(P \cdot s) \cap (P' \cdot s)\big| \;=\; |A\backslash U/B|,$$

the number of points where the two factor-orbits cross. Here $U = \mathrm{Stab}_G(s)$ is the isotropy group of s and $A = U \cap P$, $B = U \cap P'$ are the isotropies of the two factors. That $h(s)$ is exactly the double-coset count $|A\backslash U/B|$, and that $h(s) = 1$ if and only if the composition is *aligned* (the two orbits meet only at s), was established in [6]. Misalignment — $h(s) > 1$ — is the obstruction to

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treating the composite as a single well-defined move; it is the container-theoretic signature of a reentrancy bug.

The question. The count $h(s)$ is a combinatorial quantity. Is it *cohomological*? Emergent holonomy sits in a family of results where the obstruction to a clean composition is an H^2-class [8, 9], and it is natural to expect the same here: that alignment should be witnessed by the vanishing of some higher cohomology class attached to the pair (A, B) inside U. A concrete form of the expectation, due to a collaborator, was that misalignment should surface as a nonzero class in $\mathrm{Ext}^2_{kU}(k[U/A], k[U/B])$ or higher.

The result. It does not. The holonomy invariant is representation-theoretic, but it lives entirely in degree zero, and the higher tower is not merely uninformative — it is identically zero.

Let k be a field, $M = k[U/A]$ and $N = k[U/B]$ the permutation modules on the coset spaces. The classical Mackey–Eckmann–Shapiro decomposition of Ext between permutation modules (Theorem 9) reads

$$\mathrm{Ext}^n_{kU}(k[U/A], k[U/B]) \;\cong\; \bigoplus_{u \in A\backslash U/B} \mathrm{H}^n\big(A \cap uBu^{-1}; k\big), \qquad n \geq 0.$$

In degree zero every summand is $\mathrm{H}^0(H; k) = k$, so the rank of Ext^0 is the number of double cosets. Combined with [6] this is the dictionary:

Theorem 1 (Dictionary, degree zero; Corollary 11)**.** *In the emergent-holonomy setting,*

$$h(s) \;=\; |A\backslash U/B| \;=\; \dim_k \mathrm{Hom}_{kU}(k[U/A], k[U/B]) \;=\; \dim_k \mathrm{Ext}^0_{kU}(k[U/A], k[U/B]).$$

The point of this note is the next statement, which is what makes the degree-zero location *forced* rather than incidental.

Theorem 2 (Degree-zero concentration; Theorem 12)**.** *In the emergent-holonomy setting, for* every $u \in U$ *the intersection* $A \cap uBu^{-1}$ *is trivial. Consequently*

$$\mathrm{Ext}^n_{kU}(k[U/A], k[U/B]) \;\cong\; \begin{cases} k^{\,h(s)} & n = 0, \\ 0 & n \geq 1, \end{cases}$$

independently of $\mathrm{char}\, k$ *and of whether the composition is aligned.*

The mechanism is one line: $A \subseteq P$ and $uBu^{-1} \subseteq uP'u^{-1}$, so $A \cap uBu^{-1} \subseteq P \cap uP'u^{-1} = \{e\}$ because an exact factorisation is *transverse* — every conjugate of P' meets P trivially (Lemma 4). Every Mackey summand in the decomposition is then the cohomology of the trivial group, which is k in degree 0 and 0 above.

What this corrects, and why it matters. The working hypothesis — "a higher class detects alignment" — is false, and Theorem 2 says precisely why. The alignment content is the *rank of* Ext^0: aligned $\iff$ $h(s) = 1$ $\iff$ Ext^0 is one-dimensional. The higher tower carries none of it. Our verification (Section 5) includes a decisive witness, computation **W2**: an exact factorisation of S_4 with $h(s) = 2 > 1$ (misaligned) whose entire higher Ext-tower is nevertheless zero. A higher class cannot be detecting what a misaligned example leaves invisible.

This also settles the collaborator's Ext^2 conjecture in a clean way: it was right in degree 0 (where $h = \dim \mathrm{Ext}^0$) and structurally void in every higher degree, because transversality leaves no higher Ext for a class to inhabit [11]. The same transversality explains a twist-stability phenomenon (Corollary 14): tensoring N with any character leaves the tower unchanged, so an "orientation line" cannot revive a higher obstruction.

For the wider programme the payoff is operational. Emergent holonomy of an unprotected orchestration now carries a representation-theoretic signature, $h(s) = \dim_k \mathrm{Hom}_{kU}(k[U/A], k[U/B])$, computable as a single space of equivariant maps — or, with no homological algebra at all, as a double-coset count — *together with a guarantee* (Theorem 2) that no deeper Ext invariant can conceal the obstruction. Detection is a degree-zero computation, and there is nothing further to audit.

On novelty. The decomposition of Theorem 9 is classical: it is the Mackey formula for Ext between permutation modules, going back to Nakaoka, and it is standard textbook material [1, 2]. We give it in coordinates, with each ingredient cited, because the container-theory audience for whom this note is written may not have the group-cohomology backbone to hand. The contribution is the identification of $h(s)$ with $\dim \mathrm{Ext}^0$ and the degree-zero concentration theorem, together with the corrected reading they force.

Outline. Section 2 fixes notation for permutation modules, group cohomology, and the exact-factorisation setup. Section 3 assembles the three classical ingredients into the Mackey Ext-decomposition. Section 4 proves the dictionary and the degree-zero concentration, and derives twist-stability. Section 5 reports the independent numerical verification. Section 6 steps outside the holonomy locus to place the vanishing inside a single abelian factorisation $\mathrm{Ext} = h \cdot \mathrm{H}^*(A \cap B)$, worked out on V_4 and read geometrically through support varieties. Section 7 draws the operational conclusion and separates emergent holonomy from the superficially similar higher towers that arise for unrelated subgroups.

2. Preliminaries

Throughout, G is a finite group, k a field of characteristic p (the interesting case is $p \mid |G|$), and all modules are *left* modules over the group algebra kG. For $g \in G$ and $B \leq G$ write ${}^gB := gBg^{-1}$.

2.1. **Induction, restriction, permutation modules.** For $H \leq G$ and a kH-module W, induction and restriction are

$$\mathrm{Ind}_H^G W \;=\; kG \otimes_{kH} W, \qquad \mathrm{Res}_H Y \;=\; Y \text{ regarded as a } kH\text{-module.}$$

The *permutation module* on the left cosets G/H is $k[G/H] = \mathrm{Ind}_H^G k$, the free k-module on G/H with G permuting the basis. We write $M = k[G/A] = \mathrm{Ind}_A^G k$ and $N = k[G/B] = \mathrm{Ind}_B^G k$.

2.2. **Group cohomology.** For $H \leq G$ we write $\mathrm{H}^n(H; k) := \mathrm{Ext}^n_{kH}(k, k)$ for group cohomology with trivial coefficients. We use two standard facts. First, if $p \nmid |H|$ then kH is semisimple (Maschke), so $\mathrm{H}^{\geq 1}(H; k) = 0$; only p-divisible subgroups contribute a higher tower. Second, $\mathrm{H}^n(\{e\}; k) = k$ for $n = 0$ and 0 for $n \geq 1$.

2.3. **Double cosets.** For $A, B \leq G$, $A\backslash G/B$ denotes the set of double cosets AgB. A choice of representatives g is understood; different representatives of one double coset yield conjugate intersection subgroups $A \cap {}^gB$, hence cohomologically isomorphic summands, so every double-coset-indexed sum below is well defined.

2.4. **The emergent-holonomy setup.** An *exact factorisation* (Zappa–Szép product) of G is a pair of subgroups $P, P' \leq G$ with $P \cap P' = \{e\}$ and $PP' = G$; equivalently every $g \in G$ factors uniquely as $g = pp'$ with $p \in P$, $p' \in P'$. Fix a G-set S and $s \in S$, and set

$$U := \mathrm{Stab}_G(s), \qquad A := \mathrm{Stab}_P(s) = U \cap P, \qquad B := \mathrm{Stab}_{P'}(s) = U \cap P'.$$

Note $A \subseteq P$ and $B \subseteq P'$. The *emergent holonomy* at s is the crossing count of the two factor-orbits,

$$h(s) := \big|(P \cdot s) \cap (P' \cdot s)\big|.$$

We take as given the following identity, proved in [6].

Proposition 3 (Meeting points [6]). *$h(s) = |A\backslash U/B| = |U|/(|A|\,|B|)$, and $h(s) = 1$ if and only if the composition is aligned, i.e. $(P\cdot s)\cap(P'\cdot s) = \{s\}$.*

The single structural fact about exact factorisations that we need is transversality.

Lemma 4 (Transversality [6]). *Let $G = P\cdot P'$ be an exact factorisation. Then $P\cap {}^{g}P' = \{e\}$ for every $g\in G$.*

Proof. Suppose $x \in P\cap gP'g^{-1}$, so $x = p\in P$ and $x = gp'g^{-1}$ for some $p'\in P'$. Write $g = qq'$ with $q\in P$, $q'\in P'$ (unique factorisation). Conjugating x by q^{-1} gives, on the one hand, $q^{-1}xq = q^{-1}pq\in P$, and on the other, $q^{-1}xq = q^{-1}(qq')p'(qq')^{-1}q = q'p'q'^{-1}\in P'$. Hence $q^{-1}xq\in P\cap P' = \{e\}$, so $x = e$. ☐

Remark 5. Lemma 4 is the only place exactness is used. It says the factor P is transverse to *every* conjugate of P', not merely to P' itself — this global transversality is exactly what will kill the higher Ext-tower in Section 4.

3. The Mackey decomposition of Ext between permutation modules

This section assembles the classical decomposition. It rests on three ingredients, each a standard lemma stated with a one-line justification; the references are Benson [1, §2.8, §3.3] and Brown [2, III.5–III.6]. Readers fluent in group cohomology may skip to Theorem 9.

3.1. **Eckmann–Shapiro.**

Lemma 6 (Eckmann–Shapiro). *For $H\le G$, any kH-module W and any kG-module Y,*

$$\operatorname{Ext}^n_{kG}\big(\operatorname{Ind}_H^G W,\ Y\big)\ \cong\ \operatorname{Ext}^n_{kH}\big(W,\ \operatorname{Res}_H Y\big),\qquad n\ge 0.$$

Proof. Choosing coset representatives $G = \bigsqcup_i g_iH$ exhibits kG as a free right kH-module, so $\operatorname{Ind}_H^G = kG\otimes_{kH}(-)$ is exact; and it sends free modules to free modules ($\operatorname{Ind}_H^G kH = kG$), hence preserves projectives. Thus a projective resolution $P_\bullet\to W$ over kH induces a projective resolution $\operatorname{Ind}_H^G P_\bullet\to\operatorname{Ind}_H^G W$ over kG, and the adjunction $\operatorname{Ind}_H^G\dashv\operatorname{Res}_H$ gives a natural isomorphism of complexes $\operatorname{Hom}_{kG}(\operatorname{Ind}_H^G P_\bullet, Y)\cong\operatorname{Hom}_{kH}(P_\bullet,\operatorname{Res}_H Y)$. Take cohomology. ☐

Specialising $H = A$, $W = k$, $Y = N$ gives

$$\operatorname{Ext}^n_{kG}(k[G/A], N)\ \cong\ \operatorname{Ext}^n_{kA}(k,\operatorname{Res}_A N)\ =\ \mathrm{H}^n(A;\operatorname{Res}_A N). \tag{1}$$

3.2. **Mackey.**

Lemma 7 (Mackey decomposition). *As kA-modules,*

$$\operatorname{Res}_A^G\operatorname{Ind}_B^G k\ \cong\ \bigoplus_{g\in A\backslash G/B}\operatorname{Ind}^A_{A\cap {}^gB} k.$$

Proof. Decompose $G = \bigsqcup_g AgB$ into double cosets. As an (A,B)-set each double coset is $AgB\cong A\times_{A\cap {}^gB} gB$, so the permutation A-set underlying $\operatorname{Res}_A(G/B)$ is $\bigsqcup_g A/(A\cap {}^gB)$. Linearising gives the claim. (In general Mackey the g-th summand carries a twist ${}^g(-)$ of the coefficient module; for the trivial coefficient k the twist is trivial.) ☐

Because $\mathrm{H}^n(A;-)$ commutes with finite direct sums in the coefficient module, (1) and Lemma 7 combine to

$$\operatorname{Ext}^n_{kG}(k[G/A], k[G/B])\ \cong\ \bigoplus_{g\in A\backslash G/B}\mathrm{H}^n\big(A;\operatorname{Ind}^A_{A\cap {}^gB} k\big). \tag{2}$$

3.3. Shapiro for cohomology.

Lemma 8 (Coinduction form)**.** *For $H \le A$,* $\quad \mathrm{H}^n(A; \mathrm{Ind}_H^A k) \cong \mathrm{H}^n(H; k)$.

Proof. For finite index, induction and coinduction coincide: $\mathrm{Ind}_H^A \cong \mathrm{Coind}_H^A$, both $k[A/H]$ as a space. Now Coind_H^A is right adjoint to the exact functor Res_H, which preserves projectives (kA is free over kH), so the derived adjunction gives $\mathrm{Ext}^n_{kA}(k, \mathrm{Coind}_H^A k) \cong \mathrm{Ext}^n_{kH}(\mathrm{Res}_H k, k) = \mathrm{Ext}^n_{kH}(k, k) = \mathrm{H}^n(H; k)$. □

3.4. The decomposition.

Substituting Lemma 8 (with $H = A \cap {}^gB$) into (2):

Theorem 9 (Mackey–Eckmann–Shapiro; classical, cf. [1, §3])**.** *For a finite group G, subgroups $A, B \le G$, and any field k,*

$$\mathrm{Ext}^n_{kG}(k[G/A], k[G/B]) \;\cong\; \bigoplus_{g \in A\backslash G/B} \mathrm{H}^n\big(A \cap gBg^{-1}; k\big), \qquad n \ge 0,$$

naturally in n, the sum over any set of double-coset representatives.

We claim no novelty for Theorem 9; it is the Mackey formula for Ext of permutation modules (Nakaoka; see [1, §3]). Its degree-zero part is worth recording separately, since it needs no resolution at all.

Corollary 10. $\dim_k \mathrm{Hom}_{kG}(k[G/A], k[G/B]) = |A\backslash G/B|$.

Proof. Take $n = 0$ in Theorem 9: every summand is $\mathrm{H}^0(-; k) = k$. Directly, $\mathrm{Hom}_{kG}(k[G/A], N) = (\mathrm{Res}_A N)^A$, and the A-fixed points of the permutation module $k[G/B]$ are spanned by the A-orbit sums on G/B, i.e. one basis vector per double coset AgB. □

4. The dictionary and degree-zero concentration

We now specialise to the emergent-holonomy setup of §2.4, applying the machinery of Section 3 *with G replaced by the stabiliser group U* and the subgroups $A = U \cap P$, $B = U \cap P'$.

4.1. The dictionary.

Corollary 11 ($h(s) = \dim \mathrm{Ext}^0$)**.** *In the emergent-holonomy setting,*

$$h(s) \;=\; |A\backslash U/B| \;=\; \dim_k \mathrm{Hom}_{kU}(k[U/A], k[U/B]) \;=\; \dim_k \mathrm{Ext}^0_{kU}(k[U/A], k[U/B]).$$

Proof. The outer equality is Proposition 3; the inner two are Corollary 10 applied over U (with $\mathrm{Ext}^0 = \mathrm{Hom}$). □

So the double-coset invariant one hoped Ext would see *is* Ext — in degree zero. The remaining question is whether it is *only* in degree zero.

4.2. The higher tower vanishes.

Theorem 12 (Degree-zero concentration)**.** *In the emergent-holonomy setting, for every $u \in U$ the intersection $A \cap uBu^{-1}$ is trivial. Consequently, over U,*

$$\mathrm{Ext}^n_{kU}(k[U/A], k[U/B]) \;\cong\; \bigoplus_{u \in A\backslash U/B} \mathrm{H}^n(\{e\}; k) \;=\; \begin{cases} k^{\,h(s)} & n = 0, \\ 0 & n \ge 1, \end{cases}$$

independently of $\mathrm{char}\, k$ *and of alignment.*

Proof. Since $A \subseteq P$ and $uBu^{-1} \subseteq uP'u^{-1}$,

$$A \cap uBu^{-1} \;\subseteq\; P \cap uP'u^{-1} \;=\; \{e\}$$

by Lemma 4, applicable because $u \in U \subseteq G$. Every Mackey summand of Theorem 9 (over U) is therefore $\mathrm{H}^n(\{e\}; k)$, which is k for $n = 0$ and 0 for $n \geq 1$. Summing over the $h(s) = |A\backslash U/B|$ double cosets gives the result. □

Remark 13 (The corrected reading)**.** Three consequences deserve to be stated plainly.

(i) The holonomy count is cohomological, but as a *degree-zero rank* $h(s) = \dim \mathrm{Ext}^0$, not as a higher cohomology class. The tower is $[\,h, 0, 0, \dots]$; there is no H^2-shadow of h in it, and no higher class at all.

(ii) Alignment is read off in degree zero: aligned $\iff$ $h(s) = 1$ $\iff$ Ext^0 is one-dimensional; misaligned $\iff$ $h(s) > 1$ $\iff$ $\dim \mathrm{Ext}^0 > 1$. Computation **W2** below exhibits $h(s) = 2 > 1$ (misaligned) with the higher tower still identically zero, so the higher tower demonstrably does *not* detect alignment.

(iii) The transversality that forces the vanishing is exactly why an Ext^2-shifted-class ansatz was structurally doomed [11]: in any exact-factorisation holonomy setting Lemma 4 makes $\mathrm{Res}_A N$ free over kA, killing $\mathrm{Ext}^{\geq 1}$ — there is no Ext^2 for a class to live in. The ansatz was right in degree 0 and void above it.

4.3. **Twist-stability.** The degree-zero concentration is robust under twisting the target module by a character, which is the natural way one might try to inject an orientation into the invariant.

Corollary 14 (Twist-stability)**.** *Let L be any one-dimensional kG-module (character). Then*

$$\mathrm{Ext}^n_{kG}(k[G/A], k[G/B] \otimes L) \;\cong\; \bigoplus_{g \in A\backslash G/B} \mathrm{H}^n\big(A \cap gBg^{-1};\, \mathrm{Res}_{A\cap gBg^{-1}} L\big).$$

In the emergent-holonomy setting every $A \cap uBu^{-1} = \{e\}$, so $\mathrm{Res}\, L = k$ and the tower is unchanged by the twist; over $\mathbb{F}_2$ there is no nontrivial character at all.

Proof. Mackey with coefficients gives $\mathrm{Res}_A(N \otimes L) = \bigoplus_g \mathrm{Ind}^A_{A\cap {}^gB} \mathrm{Res}_{A\cap {}^gB}({}^gL)$; apply Lemma 6 and 8. Since L is one-dimensional, G acts on it by a character, which is conjugation-invariant, so ${}^gL = L$. In the holonomy setting each intersection is trivial, so the restricted coefficient is k. □

Thus an "orientation line" cannot revive a higher obstruction: the invariant is the rank of Ext^0 and nothing about a character changes it.

5. Verification

Because Theorem 9 is assembled from several classical lemmas and the whole point is the numerical value of the tower, we verified it independently rather than trusting the assembly. The engine (`scratch/ext-mackey-general/`) implements its own exact linear algebra over $\mathbb{F}_p$ — no computer algebra system — and computes each side of Theorem 9 by a different route:

- **Left side:** a genuine minimal free resolution of $k[G/A]$ over kG, then $\mathrm{H}^n\,\mathrm{Hom}_{kG}(F_\bullet, k[G/B])$.
- **Right side:** the Mackey double cosets $A\backslash G/B$, with each $\mathrm{H}^*(A \cap gBg^{-1}; k)$ computed from its own kH-resolution.

All 17 cases agree, and $\dim \mathrm{Ext}^0 = |A\backslash G/B|$ in every case. The direct left-side computation also pins the Mackey index as $A \cap gBg^{-1}$ (rather than $gAg^{-1} \cap B$). A representative selection:

Case	p	$\mathrm{Ext}^{0,1,2,3}$	note
$V_4,\ A = \langle a\rangle, B = \langle b\rangle$	2	$[1, 0, 0, 0]$	transverse, 1 dcoset
$V_4,\ A = B = \langle a\rangle$	2	$[2, 2, 2, 2]$	2 dcosets, each $\cap = \mathbb{Z}/2$
$S_3,\ A = B = \langle(12)\rangle$	2	$[2, 1, 1, 1]$	mixed tower
$A_4,\ A = B = V_4$	2	$[3, 6, 9, 12]$	each $\cap = V_4$, $\mathrm{H}^n(V_4) = n{+}1$
$D_4,\ A = B = \langle\text{refl}\rangle$	2	$[3, 2, 2, 2]$	non-abelian 2-group
$S_4,\ A = B = \langle(12)\rangle$	2	$[7, 2, 2, 2]$	7 dcosets
$\mathbb{Z}/3,\ A = B = \{e\}$	3	$[3, 0, 0, 0]$	regular, odd p
W1: $S_3{=}A_3{\cdot}\langle(12)\rangle$, $U{=}\langle(23)\rangle$	2	$[2, 0, 0, 0]$	$\mathrm{Ext}^0 = h = 2$
W2: $S_4{=}A_4{\cdot}\langle(12)\rangle$, $U{=}S_3, A{=}A_3, B{=}\{e\}$	2	$[2, 0, 0, 0]$	$h = 2{>}1$, tower $\equiv 0$

The A_4/V_4 row is the strongest single stress-test of the assembly: the direct kG resolution has Betti numbers $[1, 2, 3, 4, 5]$ and independently reproduces the elementary-abelian tower $[3, 6, 9, 12]$. The two rows below the rule are the holonomy witnesses. **W1** confirms $\mathrm{Ext}^0 = h = 2$ with vanishing higher tower. **W2** is the decisive one for Remark 13(ii): it realises a misaligned composition ($h = 2 > 1$) whose higher Ext-tower is identically zero — a higher class cannot be the alignment obstruction, because here alignment fails while every higher group is zero.

6. The full dichotomy: $\mathrm{Ext} = h \cdot \mathrm{H}^*(A \cap B)$ over an abelian group

Theorem 12 delivers the holonomy tower $[\,h, 0, 0, \ldots\,]$ by trivialising every Mackey intersection. It is worth seeing this vanishing not as an isolated fact but as one corner of a single formula. When the ambient group is *abelian*, Theorem 9 collapses to a product of two independent invariants — a degree-independent multiplicity and a degree-graded shape — and the degree-zero concentration is exactly the case where the shape degenerates.

6.1. **The factorisation.** Let G be a finite abelian group. Then ${}^gB = B$ for every g, so all the Mackey summands of Theorem 9 share one intersection $A \cap B$; moreover $AB \leq G$ is a subgroup and each double coset $AgB = g(AB)$ is a coset of it, so

$$|A\backslash G/B| = [G : AB] = \frac{|G|\,|A \cap B|}{|A|\,|B|}.$$

Writing $h := [G : AB]$ for this meeting count, Theorem 9 reads:

Corollary 15 (Abelian factorisation)**.** *For a finite abelian group G, subgroups $A, B \leq G$, and any field k,*

$$\mathrm{Ext}^n_{kG}(k[G/A], k[G/B]) \;\cong\; \mathrm{H}^n(A \cap B; k)^{\oplus h}, \qquad h = [G : AB], \quad n \geq 0,$$

so that

$$\dim_k \mathrm{Ext}^n_{kG}(k[G/A], k[G/B]) \;=\; \underbrace{[G : AB]}_{\textit{meeting count } h} \;\cdot\; \underbrace{\dim_k \mathrm{H}^n(A \cap B; k)}_{\textit{shape of the overlap}}.$$

Proof. Abelian G makes every ${}^gB = B$, so each of the $|A\backslash G/B|$ summands of Theorem 9 is $\mathrm{H}^n(A \cap B; k)$; the count is $[G : AB]$ as computed above. □

The content is a separation of variables. The meeting count h is a *degree-independent multiplicity*: it scales every degree of the tower equally, and it is the same combinatorial number $|A\backslash G/B|$ that measures holonomy. The cohomology $\mathrm{H}^*(A \cap B; k)$ is the *per-degree shape*: it alone decides whether a tower dies after degree zero, stays constant, or grows. Neither factor is new — this is Theorem 9 read off in the abelian case — but the factored form is the right way to see the dichotomy.

A	B	$A \cap B$	h	$\dim \mathrm{Ext}^n$, $n = 0..6$	regime
$\langle a \rangle$	$\langle a \rangle$	$\langle a \rangle$	2	$[2, 2, 2, 2, 2, 2, 2]$	non-transverse, constant
$\langle a \rangle$	V_4	$\langle a \rangle$	1	$[1, 1, 1, 1, 1, 1, 1]$	one $\mathrm{H}^*(\mathbb{Z}/2)$ copy
$\langle a \rangle$	$\langle b \rangle$	$\{e\}$	1	$[1, 0, 0, 0, 0, 0, 0]$	transverse
V_4	V_4	V_4	1	$[1, 2, 3, 4, 5, 6, 7]$	$= \mathrm{H}^*(V_4)$

TABLE 1. The four permutation-module Ext towers over $\mathbb{F}_2 V_4$, from Corollary 15. The multiplicity h is a uniform vertical scaling: rows one and two are the same shape $[1, 1, 1, \dots]$ at $h = 2$ and $h = 1$. The transverse row is the aligned holonomy case of Theorem 12; the growing row is $\mathrm{H}^*(V_4)$ itself.

6.2. **Degree-zero concentration is the transverse corner.** The vanishing theorem is now the degenerate value of the shape factor.

Corollary 16 (Transverse = trivial overlap)**.** *In Corollary 15, if $A \cap B = \{e\}$ then $\mathrm{H}^n(A \cap B; k)$ is k for $n = 0$ and 0 for $n \geq 1$, so the tower is $[h, 0, 0, \dots]$. If instead $A \cap B \neq \{e\}$ and $p \mid |A \cap B|$, the tower has a nonzero higher part.*

For an abelian group an exact factorisation $G = A \cdot B$ is precisely a pair with $A \cap B = \{e\}$ and $AB = G$, i.e. $h = 1$; Corollary 16 then returns $[1, 0, 0, \dots]$, the aligned instance of Theorem 12. So over an abelian group the holonomy vanishing and the surviving non-transverse towers are two values of one formula, switched by whether the overlap $A \cap B$ is trivial.

Remark 17. Theorem 12 is *not* subsumed by Corollary 15: for a non-abelian stabiliser U the vanishing needs $A \cap {}^u B = \{e\}$ for *every* u, which Lemma 4 supplies but which a single-intersection statement does not. In the abelian examples below the two coincide, and the abelian formula is what exhibits the surviving towers alongside the vanishing one.

6.3. **The worked example: V_4.** Take $G = V_4 = \mathbb{Z}/2 \times \mathbb{Z}/2 = \langle a \rangle \times \langle b \rangle$ over $k = \mathbb{F}_2$, so $kG = \mathbb{F}_2[x, y]/(x^2, y^2)$ with $x = a - 1$, $y = b - 1$ — a local ring with a unique simple module. Here $\dim \mathrm{H}^n(\mathbb{Z}/2; \mathbb{F}_2) = 1$ in every degree and $\dim \mathrm{H}^n(V_4; \mathbb{F}_2) = n + 1$ (Poincaré series $(1 - t)^{-2}$). Corollary 15 predicts the four towers of Table 1, and a direct minimal-resolution computation confirms each in degrees 0 through 6 [11].

The four rows span the whole range of the shape factor over a single group: a shape that dies at degree zero (transverse), a shape that is constant ($\mathrm{H}^*(\mathbb{Z}/2)$, scaled by $h \in \{1, 2\}$), and a shape that grows linearly ($\mathrm{H}^*(V_4)$). Only the first is emergent holonomy; the others are what the tower does when the overlap $A \cap B$ is allowed to be nontrivial.

6.4. **Two structural readings.** The surviving tower raises two questions about *where* the class lives — one representation-theoretic, one geometric. Both were posed by a collaborator; we record the answers, the second invoking standard support-variety theory that we cite rather than reprove.

Is the surviving class isolable to a single double coset? No: it is intrinsically a sum. In the constant case $A = B = \langle a \rangle$ there are $h = 2$ double cosets, $\langle a \rangle\, e\, \langle a \rangle = \{e, a\}$ and $\langle a \rangle\, b\, \langle a \rangle = \{b, ab\}$, and Corollary 15 is a *direct* sum in which *each* coset carries an identical full copy of $\mathrm{H}^*(\langle a \rangle; \mathbb{F}_2) = [1, 1, 1, \dots]$. The observed $\dim \mathrm{Ext}^n = 2$ is $1 + 1$, not one distinguished class of dimension 2. The two summands are exchanged by the residual symmetry $b \cdot (-)$, which swaps the cosets, so although the decomposition is canonical (indexed by the double cosets) no single summand is preferred. When $h = 1$ (rows two and three of Table 1) the tower does sit in the unique coset — but that is the degenerate case, not the general one.

On which stratum does it live? On a proper one. To a kV_4-module M one attaches its *rank variety* $V_r(M) \subseteq \mathbb{P}^1$, the projectivised locus of $u = \alpha x + \beta y$ acting non-freely on M; by the theory of Carlson and Avrunin–Scott the support of $\mathrm{Ext}^*_{kG}(M, N)$ is the intersection $V_r(M) \cap V_r(N)$ [4, 5, 3]. For $M = k[V_4/\langle a\rangle]$ (dimension 2) the generator a acts trivially, so $x = a-1$ acts as the *zero* matrix, while b swaps the two cosets, so $y = b-1$ has rank 1; hence $u = \alpha x+\beta y = \beta(b-1)$ acts freely iff $\beta \neq 0$. Because x acts as the literal zero matrix this is a parametric identity over any field, giving

$$V_r\big(k[V_4/\langle a\rangle]\big) = \{(1{:}0)\} \subsetneq \mathbb{P}^1,$$

a single point — a proper, 0-dimensional stratum. The dichotomy is then a variety intersection:

- **transverse** ($A = \langle a\rangle$, $B = \langle b\rangle$): $V_r(M) = \{(1{:}0)\}$ and $V_r(N) = \{(0{:}1)\}$ are *disjoint*, so $\mathrm{Ext}^{\geq 1}$ has empty support and vanishes — the tower $[1, 0, \dots]$ of Theorem 12;
- **non-transverse** ($A = B = \langle a\rangle$): both varieties are the *same* point $\{(1{:}0)\}$, so the support is that point and $\mathrm{Ext}^{\geq 1} \neq 0$.

The surviving class is thus not a top-stratum (complexity-2, full-support) class but a complexity-1 class supported on the rank-1 stratum in the "a-direction" — consistent with its *constant* dimension $\dim \mathrm{Ext}^n = 2$ (polynomial growth of degree 0) and the periodic Betti numbers $[1, 1, 1, \dots]$ of the resolution of $k[V_4/\langle a\rangle]$. Emergent holonomy, by contrast, lives where transversality forces two such varieties apart: disjoint points, empty overlap, degree-zero tower. This is the geometric shadow of Lemma 4.

7. Conclusion

The emergent holonomy of an unprotected Zappa–Szép orchestration has a clean representation-theoretic signature:

$$h(s) \;=\; \dim_k \mathrm{Hom}_{kU}\big(k[U/A],\, k[U/B]\big),$$

the rank of the space of U-equivariant maps between the two factor-orbit permutation modules. This is a degree-zero quantity, and Theorem 12 guarantees that it is the *only* Ext-invariant of the pair: transversality of the factorisation forces the entire higher tower to vanish. Operationally, detecting misalignment is a single Hom-computation — or, with no homological algebra, a double-coset count — and one is assured that no higher Ext obstruction lurks beneath it. There is nothing deeper to audit.

This corrects the natural but mistaken expectation, drawn from the H^2-classified obstructions elsewhere in the programme [8, 9], that alignment should be detected by a higher cohomology class. It is not. The alignment content is the rank of Ext^0, and the witness **W2** shows a misaligned composition with a completely trivial higher tower. Consistently with the rigidity results of [10], there is no H^2-class for h in this tower.

A phenomenon not to conflate. The higher Ext-tower is not always zero; it vanishes here *because* of the exact factorisation. For *general* subgroups $A, B \leq G$ not arising from an exact factorisation, Theorem 9 gives a genuine higher tower $\bigoplus_g \mathrm{H}^{\geq 1}(A \cap gBg^{-1}; k)$, supported on those double cosets with $p \mid |A \cap gBg^{-1}|$; Section 6 works this out in full for abelian G, where it factors as $h \cdot \mathrm{H}^*(A \cap B)$ and emergent holonomy is the transverse corner $A \cap B = \{e\}$. But the surviving tower is a different invariant — p-divisible *overlap* between two subgroups — and it must not be read as emergent holonomy, which by Lemma 4 always sits in the transverse locus where the tower is zero.

Directions. Two questions suggest themselves. First, the trivial higher tower says the pair $(k[U/A], k[U/B])$ has $\mathrm{Res}_A N$ free over kA; it would be worth recording the resulting Ext-triviality as a structural statement about the container of factor-orbits, and asking whether it characterises exact factorisations among all subgroup pairs with a given Ext^0. Second, the surviving tower $h \cdot \mathrm{H}^*(A \cap B)$ of Section 6 is a real invariant of *overlap*, and its shape factor $\mathrm{H}^*(A \cap B)$ is explicit for abelian G; whether this graded refinement of "how badly two unrelated agents share state" has an orchestration meaning of its own — and what the shape factor becomes when U is non-abelian, where ${}^uB$ genuinely varies — is open, and would sit alongside emergent holonomy rather than inside it.

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Kodamai
Email address: neil@kodamai.com